

BEACON

Autonomous Disaster Response System — Mission Report

Scenario: WILDFIRE

Generated: 2026-05-18 19:05:06

1. MISSION OVERVIEW

Field	Value
Scenario	WILDFIRE
Location	Allen, TX — Sector 4 North Grid
Start Time	2026-05-11 14:30:00
End Time	2026-05-11 16:17:23
Duration	01h 47m 23s
Max Altitude	45.0 m (148 ft)
Battery Start	100%
Battery End	23%
Battery Used	77%
Models Active	16

2. WHAT BEACON FOUND

Survivor Candidates Detected: 3

#	Latitude	Longitude	Bearing	Confidence	Temperature
1	33.1584	-96.6735	47°	91%	38.2°C
2	33.1612	-96.6698	112°	76%	37.8°C
3	33.1557	-96.6751	223°	68%	37.1°C

Metric	Value
Structures Assessed	23
Structures Critical	7
Hazard Zones Active	4

3. PERSONNEL STATUS

Status	Count
Reported Missing	19
Unaccounted	14
Located (Thermal)	3
Recovered	2
Deceased (Confirmed)	0

4. DECISION LOG

Time	Recommendation	Outcome
14:35:22	Deploy Drone-1 to Sector 4 North — thermal candidate at bearing 047°	EXECUTED
14:52:11	Expand search radius to 500m — Mattson Zone 2 probability 61%	EXECUTED
15:18:44	Return to home — battery at 30%, estimated 4min remaining	REJECTED
15:31:02	Initiate return to home — battery at 23%	EXECUTED

5. FLIGHT DATA

Failure Prediction Events:

- 15:18:44 – CAUTION: Battery 30.0%, estimated 4.2min remaining
- 15:28:01 – WARNING: Vibration magnitude 18.4 m/s²

Active Predictive Models:

Rothermel · FARSITE · WindNinja · NOAA Weather · Mattson SAR · ShakeMap · HAZUS · TRIGRS · MOST · ALOHA · PAGER · Aftershock · Foreshock · SLOSH · HEC-RAS · USGS Live

6. DATA SOURCES

Source	Status
NOAA Weather API — api.weather.gov	ACTIVE
USGS Earthquake Feed — earthquake.usgs.gov	ACTIVE
NASA FIRMS Fire Detection — firms.modaps.eosdis.nasa.gov	ACTIVE
USGS Water Services — waterservices.usgs.gov	ACTIVE
NOAA NHC Hurricane Feed	ACTIVE
DART Buoy Network — ndbc.noaa.gov	ACTIVE

7. NOTES

Wind shift at 15:22 caused FARSITE model to update fire spread prediction. Ground teams notified of revised evacuation corridor. Two survivors extracted by Team Alpha at 15:45.